Adaptive Convolutional Subspace Reasoning Network for Few-Shot SAR Target Recognition

Haohao Ren, *Member, IEEE*, Xuelian Yu, *Member, IEEE*, Sen Liu, *Student Member, IEEE*, Lin Zou, *Member, IEEE*, Xuegang Wang, and Hao Tang

***Abstract*— Data-driven automatic target recognition (ATR) methods have become the mainstream in the synthetic aperture radar (SAR) community at this stage. However, in real SAR application scenarios, the scarcity of training samples is a common problem. Especially in military application scenarios, only a small number of samples of each type of target are usually available. In the case of limited sample availability, it is bound to bring challenges to the feature extraction model and classifier inference learning. In this article, we propose a novel method called adaptive convolutional subspace reasoning network (ACSRNet) to address few-shot SAR target recognition tasks. To be specific, we first present a dynamic-aware convolutional feature embedding network based on the siamese architecture, which can not only learn more transferable knowledge for the few-shot tasks but also dynamically adjust the convolution kernel according to the input data to extract more discriminative features. To effectively achieve target identity reasoning, we then resort to high-order information of samples, i.e., the idea of adaptive subspace learning, to develop a few-shot subspace classification module, which can online reason the identity of the unknown target through the spanning subspace of training samples. Meanwhile, a dual-loss is designed to train a feature embedding space with intraclass compactness and interclass divergence, aiming to facilitate subsequent classification. Meta-learning integrating random sampling episode way is introduced into the process of model training to realize few-shot SAR ATR tasks by emulating the human cognitive process. Experimental results on the moving and stationary target acquisition recognition (MSTAR) dataset demonstrate that the proposed method is competitive with some state-of-the-art for few-shot SAR ATR tasks.**

***Index Terms*— Automatic target recognition (ATR), convolutional neural network (CNN), synthetic aperture radar (SAR).**

I. Introduction

AS AN active microwave imaging device, synthetic aperture radar (SAR) has played a vital role in national defense and civilian fields on account of its all-weather, all-day-and-night, and strong penetration operating characteristics. Image interpretation is dedicated to mining valuable information from SAR images, which is the most important means to transform images into intelligence information. Automatic target recognition (ATR) is one of the critical techniques to realize intelligent SAR image interpretation [1], [2], [3], [4], [5]. It is essential to develop a discriminative feature embedding model and an efficient classifier in the SAR ATR tasks.

A. Prior Work

After decades of exploration and development, SAR ATR has received remarkable advancements. A variety of SAR ATR algorithms have been proposed recently. In a nutshell, these methods can be divided into three categories, i.e., template-based [6], model-based [7], and machine learning-based methods. The template-based method first offline establishes a template library through original SAR images or features extracted from SAR images, and then online reasons the identity of the unknown target by matching the best template from the template library. The attention to template-based methods has gradually declined in the field of SAR ATR due to the inefficiency of online matching. The model-based method is developed by modeling the physical or conceptual properties of the target. To put it simply, model-based methods first predict a set of hypotheses of the target characteristics such as its category, pose, and so forth for the SAR target through the predefined model, and then compare the predicted features with the real features to reason the identity of the target.

The machine learning-based method is committed to learning a low-dimensional feature embedding space and classifier from a great volume of training samples. In the last few decades, large quantities of machine learning-based ATR methods have been put forward in succession. To name just a few, Liu et al. [8] proposed a statistical model embedded with locality-preserving property to extract local discriminative features for SAR target recognition. In [9], monogenic signal analysis, sparse representation, and multitask learning are integrated to extract features and reason the identity of SAR targets. To extract more recognition information from SAR images, Cao et al. [10] presented a multiview SAR target recognition with locally adaptive dictionary learning. Dang et al. [11] exploited extreme value theory to design an incremental learning classifier for open-set SAR target recognition tasks.

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Haohao Ren, Xuelian Yu, Sen Liu, Lin Zou, and Xuegang Wang are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu 611731, China (e-mail: xlyu2012@uestc.edu.cn).

Hao Tang is with the Department of Information Technology and Electrical Engineering, ETH Zurich, 8092 Zurich, Switzerland (e-mail: hao.tang@vision.ee.ethz.ch).

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What these three ATR methods have in common is that the target template, model, and classifier are designed in advance. Moreover, the feature extraction and classification procedure of these methods are independent of each other, which evidently increases the complexity of an ATR system. Most importantly, these methods rely heavily on hand-crafted features, so that they generally perform poorly in most challenging ATR scenarios.

In late years, as a data-driven method in an end-to-end learning way, deep learning has received extensive attention in the field of SAR ATR [12]. Up to now, there are quite a few deep learning-based ATR methods in the SAR community. For instance, Chen et al. [13] designed an all-convolutional network structure, whose trainable parameters can be dramatically reduced by removing fully connected layers, thereby alleviating the overfitting phenomenon of the model. Pei et al. [14] developed a parallel fusion neural network to aggregate useful information from multiview SAR images, and achieved outstanding recognition performance on the moving and stationary target acquisition recognition (MSTAR) benchmark dataset. To extract discriminative features in various complex application scenarios, a multiscale convolutional capsule network structure was proposed for SAR target recognition tasks [15]. Liu et al. [16] designed a new feature fusion framework to combine scattering and deep image features for SAR target recognition tasks. Zeng et al. [17] proposed a multistream feature fusion method based on complex-valued operation, which effectively utilizes the phase information of the target. Dong et al. [18] developed a hierarchical receptive architecture based on a convolutional neural network (CNN) for target recognition in SAR images. Nevertheless, numerous methods perform well only when there are enough training samples, and their performances will degrade severely in the face of the scarcity of SAR training images.

In most realistic SAR applications, acquiring a great many SAR images is costly or even impossible. To realize SAR target recognition with limited training samples, a rotation awareness network was developed, which can extract the rotation-equivariant features instead of the rotation-invariant features to reduce the demand for data [19]. Lin et al. [20] presented a deep convolutional highway unit network to achieve SAR target recognition in the case of limited training samples. Guan et al. [21] put forward a two-stage multiscale densely connected CNN to implement SAR target recognition with limited training samples. Moreover, transfer learning has been widely applied to solve the problem of SAR target recognition with limited training samples [22], [23]. These existing works can alleviate the requirement of training samples for SAR target recognition tasks to some extent. However, these methods still require at least dozens of training samples for each category of target. In the case of some application scenarios such as military surveillance, noncooperative target recognition, and so on, where only relatively few samples are available for each category of the target, these methods are no longer effective.

In general, a task with only a few training samples is termed as a few-shot task. As for the problem of few-shot target recognition, inspired by the process of human cognition, a variety of theories and techniques have been proposed in the field of computer vision. Broadly speaking, these few-shot approaches can be categorized into three mainstreams: metric-based [24], [25], [26], [27], optimization-based [28], [29], and memory-based methods [30], [31]. In contrast, the research on few-shot SAR target recognition is still in its infancy [32], [33], [34], [35], [36]. Yang et al. [37] proposed a mixed loss graph attention network (MGA-Net) to promote the performance of few-shot SAR target recognition. Liu et al. [38] presented a bi-similarity prototypical network with capsule-based embedding (BSCapNet) to realize few-shot SAR target recognition. Wang et al. [39] proposed a multitask representation learning network to solve the problem of few-shot SAR target recognition. Overall, there are still considerable problems to be addressed for few-shot SAR target recognition.

B. Contributions

In order to realize few-shot SAR target recognition more effectively, this article proposes an adaptive convolutional subspace reasoning network (ACSRNet). The contributions of this article are as follows.

1) Considering that SAR images are sensitive to variation in radar view, a dynamic-aware convolutional feature embedding network based on the siamese architecture is presented, which can not only extract more transferable knowledge in the training stage, but also dynamically adjust the convolution kernel according to the input SAR images in the test stage to extract discriminative features for few-shot learning tasks.
2) In view of the fact that the unique SAR imaging characteristics lead to the weak prototype representation ability and the fact that it is difficult to accurately obtain the prototype of the target under the condition of limited samples, we abandon the traditional idea of prototype classification and instead resort to high-order information of samples, i.e., the idea of adaptive subspace learning, to develop a subspace classification module for the first time, which can robustly reason the identity of the target according to the projection distance between the sample to be recognized and the spanning subspace of training samples.
3) A dual-loss, i.e., the weighted sum of the embedding loss and classification loss, is designed to promote the intraclass compactness and interclass divergence in the feature embedding space, which is conducive to classifier learning.
4) Extensive experiments conducted on the MSTAR benchmark dataset manifest that the proposed method can achieve better performance than some state-of-the-art.

The remainder of this article is organized as follows. Section II elaborates each key component of the proposed ACSRNet. In Section III, extensive experiments on the MSTAR benchmark dataset are first carried out, and then qualitative and quantitative analyses are presented in detail. Finally, we conclude this article in Section IV.

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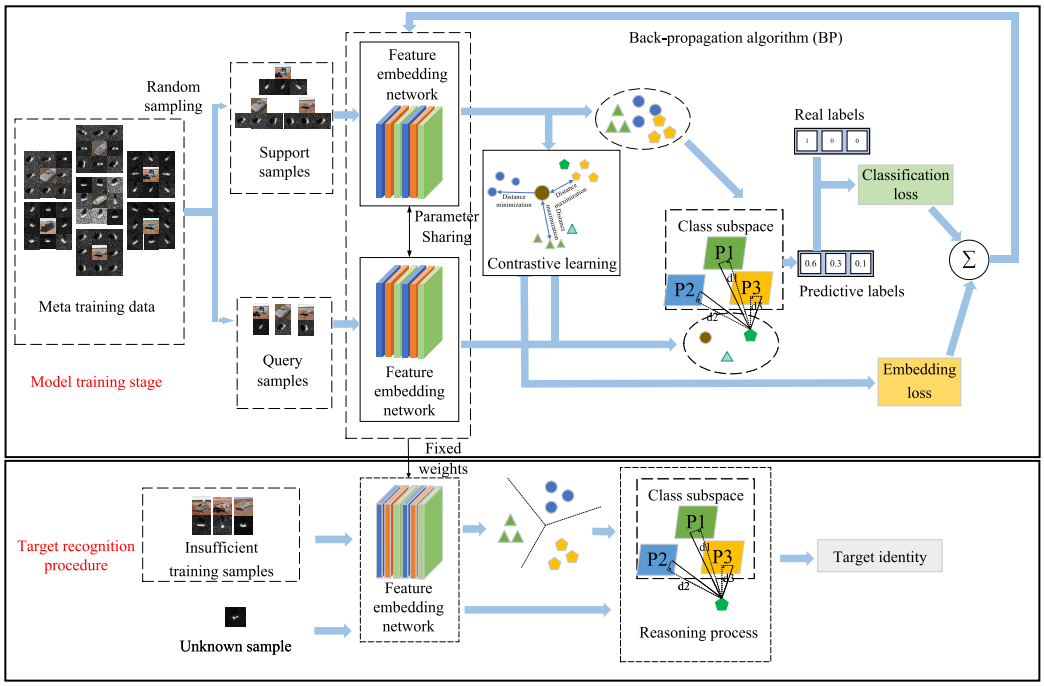


Fig. 1. Target recognition procedure of the proposed method.

II. Adaptive Convolutional Subspace Reasoning Network

The target recognition procedure of our proposed method is illustrated in Fig. 1, which consists of two stages, i.e., the model training and the target reasoning. Simply put the first stage resorts to the idea of meta-learning to learn transferrable knowledge that can be generalized to few-shot tasks from meta-training tasks. The second stage aims to effectively realize few-shot SAR target recognition with the help of the transferrable knowledge obtained in the first stage. In the following, the key components of the proposed method are elaborated.

A. Feature Embedding Module

As can be seen from Fig. 1, the feature embedding network is one of the vital components of the proposed method. Its function is to transform the high-dimensional SAR image from the original image domain to a low-dimensional feature domain that is liable to discriminate samples. Different from the traditional supervised learning model, the model in the few-shot learning task needs to learn to understand the similarities and differences between different things. As a rule, the basic idea for solving few-shot tasks is to learn a similarity measurement function. For this purpose, the feature embedding network based on siamese architecture is established to learn the similarity between the two SAR images in this article. Simultaneously, the idea of contrastive learning is injected into the entire training process of the feature embedding network, so as to learn a discriminative feature embedding space with both intraclass compactness and interclass divergence. The detailed network architecture is drawn in Fig. 2.

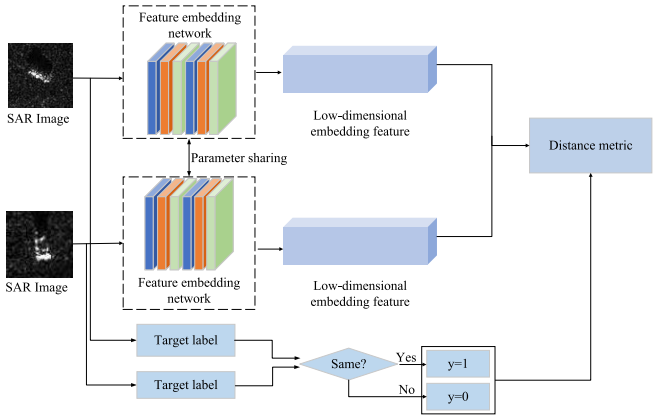


Fig. 2. Feature embedding network architecture.

In the model training stage, SAR images are input into the feature embedding network in pairs, and the category relationship between two images is indicated by an indicator y. If the two SAR images come from the same type of target, then $y = 1$, otherwise $y = 0$. Once $y = 1$, the distance between two SAR images is minimized so that the feature distribution of two SAR images is as compact as possible. Yet $y = 0$, the distance between two SAR images is maximized to enhance the degree of dispersion between different classes.

Considering that SAR images are sensitive to variation in radar view, different SAR images of the same target appear different characteristics. In response to this problem, we argue that static convolution operation may not be able to effectively extract the discriminative features of different SAR images. Accordingly, this article introduces dynamic-aware convolution operation in the feature embedding network, which can

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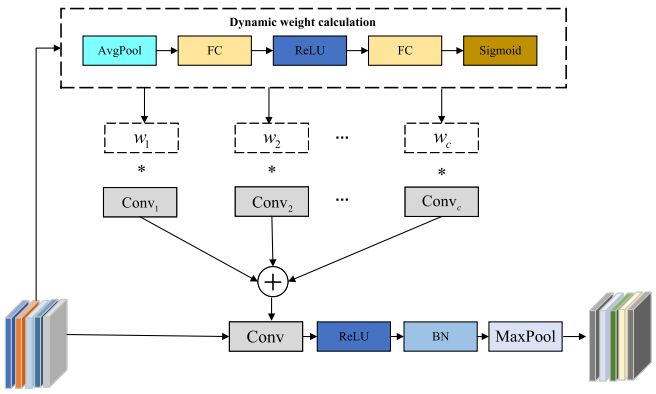


Fig. 3. Dynamic-aware convolution operation.

dynamically learn discriminative features according to the input SAR images. Concretely, each convolution layer no longer uses the same convolution kernel for all input images, but adopts multiple different convolution kernels to dynamically adapt to different SAR images. By doing so, the feature extraction network can dynamically adjust network parameters for different SAR images to mine the discriminative features of the target. To put it simply, in the dynamic-aware convolution layer, the convolution kernel is a function of the input data, similar to some previous works [40], [41], [42], [43]. Mathematically, the dynamic-aware convolution layer can be expressed as

$$\begin{aligned} \mathbf{y} &= \sigma(\mathbf{W}^T(\mathbf{x})\mathbf{x} + \mathbf{b}(\mathbf{x})) \\ \mathbf{W}(\mathbf{x}) &= \sum_{p=1}^{c} \omega_p(\mathbf{x})\mathbf{W}_p \\ \mathbf{b} &= \sum_{p=1}^{c} \omega_p(\mathbf{x})\mathbf{b}_p \\ \text{s.t. } 0 \le \omega_p \le 1, &\quad \sum_{p=1}^{c} \omega_p = 1 \end{aligned} \tag{1}$$

where $\sigma(\cdot)$ represents the activation function, T denotes transpose operation, ω_p is the dynamic weight for the pth convolution kernel, c is the total number of convolution kernel, and $\mathbf{W}_p$ and $\mathbf{b}_p$ are the coefficient matrix and bias of the pth convolution kernel, respectively.

The procedure of dynamic-aware convolution operation in the siamese feature embedding network is shown in Fig. 3. One can see that dynamic-aware convolution operation is composed of dynamic weight calculation, convolution operation, and convolution aggregation. Compared to the traditional convolution layer, the dynamic weight calculation in the dynamic-aware convolution layer is an extra operation. In this article, we adopt the idea of SENet [44] to calculate the dynamic weight ω_p $(p = 1, \ldots, c)$ of the pth convolution kernel.

In Fig. 3, AvgPool denotes average pooling operation, FC represents the fully connected layer, ReLU is the nonlinear activation function rectified linear unit, BN denotes batch normalization operation, and Conv$_p(p = 1, \ldots, c)$ represents the pth convolution kernel in a dynamic-aware convolution layer.

B. Few-Shot Subspace Classification Module

The classic classification method for solving the problem of few-shot tasks is the nearest neighbor criteria based on the idea of prototype learning. However, owing to the unique SAR imaging characteristics, SAR images are sensitive to variations in sensor azimuth angles, the otherness between the same types of SAR images acquired at different azimuth angles is relatively large. Such a phenomenon will attenuate the representation ability of embedding features after averaging to some extent. More importantly, considering that the class distribution for each type of target cannot be learned well with limited training samples, the true prototype for each type of SAR target might not be obtained in the few-shot scenarios. In view of the foregoing, it is not a good choice to exploit the idea of prototype learning to design classification criteria for few-shot SAR target recognition tasks. Therefore, how to design an efficient and robust classifier is the key to reliable few-shot SAR target recognition.

It is argued that high-order information is preferred over low-order one for promoting the discriminative ability of the classifier in the field of pattern recognition [45], [46], [47]. Inspired by this idea, we commit to a second-order method, i.e., the idea of adaptive subspace learning, to develop a classification model for few-shot SAR target recognition. In simple terms, the low-dimensional embedding features of training SAR images are first used for subspace modeling, and then the shortest projection distance of low-dimensional embedding features of SAR images on the spanning subspace of each type of training SAR images is calculated to reason the identity of the target. Suppose that $\mathbf{U}_i = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n] \in \mathbb{R}^{D\times n}$ represents the basis vector matrix of the spanning subspace of the ith class target training samples, $\mathbf{U}_i^T\mathbf{U}_i = \mathbf{I}_n$ $(n \le D)$, D denotes the dimension of the basis vector, and n is the number of the basis vector. Usually, the basis vector of the spanning subspace of each category can be solved via the matrix decomposition method. Below, the subspace modeling process of training SAR images and the identity reasoning of the target are described in detail.

First, let $\mathbf{X} = \{\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, \ldots, \mathbf{X}_{i,K}\}$ be the support samples of the ith class SAR target in a meta-task. After feature embedding network encoding, these samples can be expressed as $\{\Phi_\theta(\mathbf{x}_{i,1}), \Phi_\theta(\mathbf{x}_{i,2}), \ldots, \Phi_\theta(\mathbf{x}_{i,K})\}$, where θ is the trainable parameters of the feature embedding network. The mean value of the ith class SAR target features is described as

$$\boldsymbol{\mu}_i = \frac{1}{K}\sum_{m=1}^{K} \Phi_\theta(\mathrm{x}_{i,m}). \tag{2}$$

After subtracting the mean value of the features, the sample features can be written as

$$\widetilde{\mathbf{X}}_i = [\Phi_\theta(\mathbf{x}_{i,1}) - \boldsymbol{\mu}_i, \Phi_\theta(\mathbf{x}_{i,2}) - \boldsymbol{\mu}_i, \ldots, \Phi_\theta(\mathbf{x}_{i,K}) - \boldsymbol{\mu}_i]. \tag{3}$$

Afterward, $\widetilde{\mathbf{X}}_i$ is subjected to singular value decomposition (SVD), namely

$$\widetilde{\mathbf{X}}_i = \mathbf{U}\Sigma\mathbf{V}^T. \tag{4}$$

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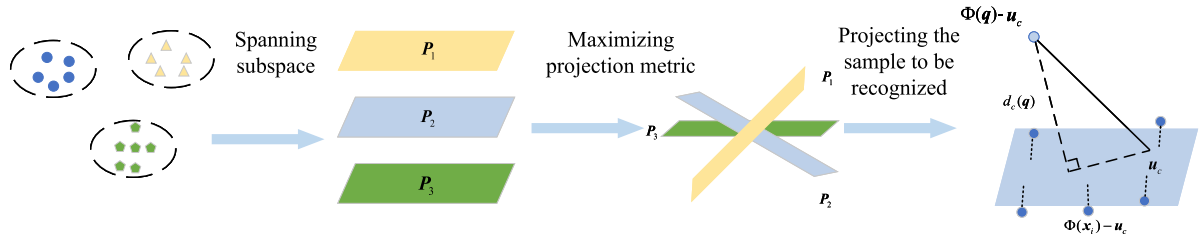


Fig. 4. Learning process of few-shot subspace classification module.

We select the first d column vectors in $\mathbf{U}$ to get the truncated matrix $\mathbf{P}_i$. Obtaining the spanning subspace $\mathbf{P}_i$ from $\widetilde{\mathbf{X}}_i$ is actually the truncated SVD.

Then, to enhance the discrimination of the spanning subspaces of different categories of targets, the distance between different subspaces should be as large as possible during model training. To this end, this article puts to use the projection metric on Grassmannian geometry to maximize the projection distance between different subspaces [48]. Specifically, given the subspace basis of $\mathbf{P}_i$ and $\mathbf{P}_j$ for two categories of SAR targets, the projection metric between two subspaces is defined as follows:

$$\delta^2(\mathbf{P}_i, \mathbf{P}_j) = \|\mathbf{P}_i\mathbf{P}_i^{\mathrm{T}} - \mathbf{P}_j\mathbf{P}_j^{\mathrm{T}}\|_F^2 = 2n - 2\|\mathbf{P}_i^{\mathrm{T}}\mathbf{P}_j\|_F^2 \tag{5}$$

where n is the dimension of the spanning subspace, and $\|\bullet\|_F$ denotes the Frobenius norm operation.

Maximizing (5) is equivalent to minimizing the following formula:

$$\arg\min \|\mathbf{P}_i^{\mathrm{T}}\mathbf{P}_j\|_F^2. \tag{6}$$

Using the above operations, the spanning subspace $\mathbf{P}_i$ of the ith class SAR target can be obtained.

Then, the low-dimensional embedding feature of the sample to be recognized then is projected onto $\mathbf{P}_i$ $(i = 1, \ldots, N)$, where N is the total number of target categories. Finally, the identity of the sample $\mathbf{q}$ to be recognized is reasoned by comparing projected distances. The projection distance [27] can be expressed as

$$d_i(\mathbf{q}) = -\|(\mathbf{I} - \mathbf{M}_i)(\Phi_\theta(\mathbf{q}) - \boldsymbol{\mu}_i)\|^2 \tag{7}$$

where $\mathbf{M}_i = \mathbf{P}_i\mathbf{P}_i^T$.

To ensure that the proposed method can make use of the backpropagation algorithm for end-to-end learning, the Softmax function is used to calculate the probability that the sample is judged to be ith class, i.e.,

$$p(i|\mathbf{q}) = \frac{\exp(d_i(\mathbf{q}))}{\sum_{i=1}^{n=N}\exp(d_i(\mathbf{q}))} \tag{8}$$

where N is the total number of target categories.

The learning process of the few-shot subspace classification module is illustrated in Fig. 4.

C. Meta-Tasks Generation

Different from the traditional deep learning model training way, the few-shot learning model with the idea of meta-learning adopts an episode-based training method [33]. In the process of episode-based training, a meta-task is a basic unit, consisting of support samples and query samples. Usually, typical few-shot learning methods require that meta-training data incorporate enough target classes to guarantee excellent generalization performance of the model. In the field of computer vision, miniImageNet and Omniglot are widely used for evaluating the effectiveness of few-shot learning methods, which include 100 and 1623 classes of targets, respectively. Nevertheless, both the target category and sample size in SAR ATR tasks are far less than those in the typical few-shot learning methods. Therefore, the typical episode-based training method is no longer effective in few-shot SAR ATR tasks. To solve this problem, this article introduces a random sampling episode method [49] to generate meta-tasks so that the model can obtain powerful generalization capabilities.

A target recognition task consisting of N types of targets with K (K is usually a very small integer) training samples for each type is termed as N-way K-shot task. Suppose that the meta-training dataset contains C types of targets, and each type of target has M samples. The meta-task generation based on random sampling episode way is mainly divided into three steps. First, N types of targets are randomly selected from the meta-training data, and then M samples are selected from the selected N types of targets. Finally, K samples are randomly selected from M samples of each type, and the remaining $(M–K)$ samples of each type are used as query samples. Fig. 5 summarizes the process of generating meta-tasks from meta-training data. By doing so, a total of $[C!/(C-N)!N!]\cdot[M!/(M-K)!K!]^N$ different support sets and query sets are generated from the meta-training dataset. Such a diverse sample set is the capability to guarantee that the model can acquire enough transferrable knowledge in the meta-training stage.

D. Network Training

To obtain a discriminative feature embedding space with intraclass compactness and interclass divergence, we adopt contrastive learning [50] as the embedding loss of the feature embedding network, which is defined as

$$L_{cl} = \frac{1}{NK(NM-NK)}\sum_{m=1}^{NK}\sum_{k=1}^{N(M-K)} y d_{m,k}^2 + (1-y)\max(\text{margin} - d_{m,k}^2, 0)^2 \tag{9}$$

where $d_{m,k} = \|\Phi_\theta(\mathbf{x}_{i,m}) - \Phi_\theta(\mathbf{x}_{j,k})\|_2$ is Euclidean distance between the low-dimensional embedding feature of the sample

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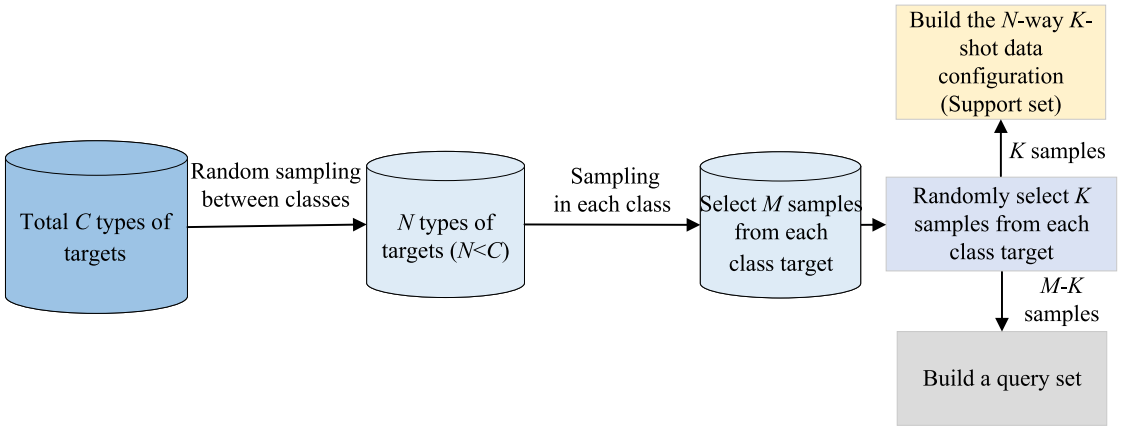


Fig. 5. Meta-task generation process.

$\mathbf{x}_{i,m}$ and $\mathbf{x}_{j,k}$, $\max(\cdot,\cdot)$ indicates the larger value between two numbers, y represents the label relationship of this pair of samples, $y = 1$ means that the two samples are from the same target, while $y = 0$ represents that the identity of the two samples is different, margin is a preset value used to restrain the margin distance between different types of samples.

Jointing (6) and (8), the classification loss of the few-shot subspace classification module is

$$L_c = -\frac{1}{N(M-K)}\sum_{i=1}^{N}\log(p_{(i|\mathbf{q})}) + \lambda\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N}\|\mathbf{P}_i^{\mathrm{T}}\mathbf{P}_j\|_F^2 \tag{10}$$

where λ is a regularization parameter.

Synthesizing (9) and (10), a dual-loss is formulated as

$$L = L_c + \gamma L_{cl} \tag{11}$$

where γ is a weighting factor to balance the importance between embedding loss and classification loss.

In this article, backpropagation through SVD [51] is required to train the whole model in an end-to-end manner. The training process of the proposed method is summarized in Algorithm 1.

Now, we analyze the computational complexity of the proposed ACSRNet. Compared with common convolution, dynamic-aware convolution contains two extra operations: dynamic weight calculation and convolution aggregation. For an input feature map with dimension $W \times H \times C_{\text{in}}$, the computational cost is $O(HWC_{\text{in}} + C_{\text{in}}^2/4 + C_{\text{in}}F/4)$, which is much less than of common convolution operation, i.e., $O(HWC_{\text{in}}C_{\text{out}}F^2)$, where F is the kernel size, and C_{out} represents the number of output channels. The computational cost of convolution aggregation is $O(VC_{\text{in}}C_{\text{out}}F^2 + VC_{\text{out}})$, where V is the number of convolution kernel with kernel size $F \times F$. In general, $F \ll H \times W$, so the computational cost of convolution aggregation is much less than that of common convolution. So, the computational cost of the feature embedding network is in the order of $O(HWC_{\text{in}}C_{\text{out}}F^2)$. The computational cost of the few-shot classification model based on deep subspace learning is $O(ND^2K)$. Therefore, the computational cost of the proposed ACSRNet is in the order of $O(\max(HWC_{\text{in}}C_{\text{out}}K^2, ND^2K))$.

Algorithm 1 Training Process of the Proposed Method

Require: meta-tasks $\{\mathcal{T}_1, \mathcal{T}_2, \cdots, \mathcal{T}_T\}$, each meta-task consists of a support set S and a query set Q

Initialize model parameters θ
for t in $\{\mathcal{T}_1, \mathcal{T}_2, \cdots, \mathcal{T}_T\}$ **do**
  for i in $\{1, \cdots N\}$ **do**
    Embed features of samples: $\Phi_\theta(\mathbf{x}_{i,m})$ $(m = 1, \cdots, K)$
    Compute the mean value of features for each class in a meta-task via Eq. (2): $\boldsymbol{\mu}_i$
    Calculate $\widetilde{\mathbf{X}}_i = [\ \Phi_\theta(\mathbf{x}_{i,1}) - \boldsymbol{\mu}_i,\ \Phi_\theta(\mathbf{x}_{i,2}) - \boldsymbol{\mu}_i,\ \cdots,\ \Phi_\theta(\mathbf{x}_{i,K}) - \boldsymbol{\mu}_i]$
    Utilize SVD method decompose $\widetilde{\mathbf{X}}_i \rightarrow [\mathbf{U}, \Sigma, \mathbf{V}^{\mathrm{T}}]$
    Truncate matrix $\mathbf{U}_{1,\ldots,n} \rightarrow \mathbf{P}_i$
    for $\mathbf{q}$ in Q **do**
      Calculate the projection distance from the query sample to the spanning subspace of the support samples via Eq. (7)
      Compute the probability that the sample belongs to class i-th target via Eq. (8)
    end for
  end for
  Calculate loss of the proposed method via Eq. (11)
  Update model parameters θ by stochastic gradient descent method
end for

Ensure: Output feature embedding model parameters: Φ_θ

III. Experimental Results

To assess the effectiveness of the proposed method, extensive experiments are performed on the publicly released MSTAR dataset. In this section, the MSTAR benchmark dataset is introduced first. Then, we give detailed experimental settings, including few-shot task generation, hyper-parameters setting, learning rate, and so forth. Finally, extensive experiments are conducted to assess the effectiveness and

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superiority of the proposed method on the MSTAR benchmark dataset.

A. Data Description

The MSTAR dataset is a benchmark dataset widely used for evaluating the performance of ATR algorithms in the SAR community, which was publicly released by the U.S. Defense Advanced Research Projects Agency (DARPA) and Air Force Research Laboratory. All SAR images in the MSTAR dataset were acquired by the U.S. Sandia National Laboratory using a 10-GHz X-band spotlight SAR sensor. Images of each target were taken at different depression angles over the full azimuth from 0° to 359°. However, due to storage, transmission, and other reasons, only partial azimuth SAR images of each type of target are available in the publicly released MSATR dataset. The resolution of the original SAR image is 0.3 × 0.3 m in both range and azimuth in the MSTAR dataset. The publicly released MSTAR dataset includes ten categories of ground military targets. Fig. 6 presents some SAR images of targets and their corresponding optical images. Considering that the original SAR image includes a mass of redundant background, all images are cropped to 64 × 64 pixels region of interest in all of the following experiments.

B. Experimental Settings

In this article, we divide the original MSTAR dataset into meta-training data and meta-test data according to the depression angle of the target. As a matter of routine [10], [23], all SAR images for seven categories of ground military targets, i.e., BMP2, BTR60, BTR70, T62, T72, D7, and ZIL131, acquired at 17° depression angle are utilized as meta-training data, and all SAR images for the other three categories of ground military targets, i.e., 2S1, BRDM2, and ZSU23/4, captured at 15°, 17°, and 30° depression angles are exploited to generate meta-test data. Table I summarizes the detailed data information of each target at different depression angles.

In the following experiments, the subspace dimension of the subspace learning-based classification module is set to $K-1$. Concerning the SAR target recognition task with only one sample for each category of the target, the SAR image is first flipped in any three directions to augment samples. Then the identity of the target is reasoned based on the spanning subspace of the augmented samples. Three parameters in (10), including margin, λ, and γ, are set to 1, 0.03, and 0.1, respectively. As a matter of routine [33], [37], [38], [49], in the following experiments, 200 samples are selected for each type of target to establish the support set and query set in the meta-training stage, and the number of samples to be recognized for each type of target in each task is set to 15. The network configuration of the feature embedding network is shown in Table II, where dynamic-aware convolution operation is represented as dconv-(number of feature maps)@(filter size), ReLU is the non-linear function, BN is batch normalization operation, and Max-pooling@(region size) is the maximum response in a region.

This article adopts the Adam optimizer [52] for optimizing the proposed method. The learning rate is set to 0.001. The proposed method is implemented with the PyTorch framework, and all experiments are carried out on a personal computer with the hardware configuration of Intel(R) Core(TM) i7-9750CPU@2.60 GHz, NVIDIA GeForce RTX2060 GPU.

TABLE I
CATEGORIES AND NUMBER OF THE TARGET UNDER DIFFERENT DEPRESSION SCENARIOS

Type	Depression	Target	Number
D_{tr}	17°	BMP2	233
		BTR60	256
		BTR70	233
		T62	299
		T72	232
		D7	299
		ZIL131	299
D_{te}	15°	2S1	274
		BRDM2	274
		ZSU23/4	274
	17°	2S1	299
		BRDM2	298
		ZSU23/4	299
	30°	2S1	288
		BRDM2	287
		ZSU23/4	288

TABLE II
FEATURE EMBEDDING NETWORK CONFIGURATION

Layer	Configuration
1	dconv-32@5×5/ReLU/BN/Max-pooling@2×2
2	dconv-64@5×5/ ReLU/BN/Max-pooling@2×2
3	dconv-128@3×3/ ReLU/ BN/ Max-pooling@2×2
4	dconv-3@5×5/softmax

C. Performance Assessment Under Different Sample Scarcity Levels

In this section, we design experiments to evaluate the recognition performance of the proposed method under different sample scarcity levels. All SAR images used in this section are acquired at 17° depression angle. To simulate few-shot SAR target recognition tasks, we generate few-shot data according to randomly sampling episode way in Section II-C. This experiment investigates four different few-shot SAR ATR tasks, i.e., the value of K is {1, 3, 5, 10} in Section II-C. Each ATR task is conducted independently for 1000 runs. In this section, four few-shot learning methods, i.e., ProtoNet [25], RelationNet [26], meta-learning SAR (MSAR) [33], and BSCapNet [38], are employed as competitors to illustrate

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Fig. 6. (Top) SAR images and (bottom) corresponding optical images. (a)–(j) BMP2, BTR70, T72, BTR60, 2S1, BRDM2, D7, T62, ZIL131, and ZSU234, respectively.

the effectiveness and superiority of the proposed method under different sample scarcity levels. Table III presents the minimum recognition accuracy, maximum recognition accuracy, average recognition accuracy, and standard deviation (STD) obtained by all methods for 1000 runs of independent experiments under different recognition tasks. Constrained by paper space, Fig. 7 presents only the statistical histogram of the recognition rate (RR) of the two methods for 1000 runs of independent experiments under different recognition tasks.

As can be seen from Table III, when there are only ten training samples for each category of the target, the RR of the proposed method can reach over 90%, which is about 5% higher than that of each competitor. In addition, the proposed method can obtain an RR of 78.38% in the one-shot scenario, while BSCapNet, as the best performance among the comparison methods, can only achieve an RR of 74%. By comparing the STD of these methods, it can be clearly observed that the STD of the proposed method is much smaller than that of each competitor, which indicates that the proposed method is more robust than these competitors. Moreover, one can see from Fig. 7 that the RR distribution of the proposed method is more concentrated than that of ProtoNet, which is consistent with experimental results reflected by STD in Table III. In summary, we can conclude that the proposed method can always perform better than this state-of-the-art under different few-shot SAR ATR tasks.

TABLE III
RECOGNITION ACCURACY OF EACH METHOD UNDER DIFFERENT SAMPLE SCARCITY LEVELS

Recognition Task	Method	Accuracy (%)			STD
		min	max	avg	
3-way 1-shot	ProtoNet	26.67	95.56	69.90	0.1001
	RelationNet	31.11	91.11	69.95	0.1051
	MSAR	20.00	100	68.43	0.1374
	BSCapNet	63.00	100	74.37	0.1079
	Ours	40.00	97.77	**78.38**	**0.0860**
3-way 3-shot	ProtoNet	48.44	97.77	79.58	0.7201
	RelationNet	51.11	95.56	79.61	0.0754
	MSAR	46.67	100	80.21	0.1011
	BSCapNet	50.00	100	82.41	0.0771
	Ours	56.67	100	**86.17**	**0.0038**
3-way 5-shot	ProtoNet	57.78	100	82.22	0.6626
	RelationNet	48.89	95.56	81.21	0.5542
	MSAR	53.33	100	83.83	0.0932
	BSCapNet	63.00	100	86.66	0.0617
	Ours	62.22	100	**87.96**	**0.0565**
3-way 10-shot	ProtoNet	71.11	100	87.61	0.5602
	RelationNet	57.78	97.78	85.33	0.0651
	MSAR	60.00	100	87.96	0.0789
	BSCapNet	63.69	100	87.92	0.0597
	Ours	75.56	100	**92.52**	**0.0028**

D. Performance Assessment at Various Depression Variations

Considering the noncooperative characteristic of the target in most real SAR application scenarios, SAR images in the support set and query set may either be from the same depression angle or different angles in an ATR task. Therefore, to further validate the effectiveness of the proposed method, we carry out five groups of evaluation experiments at different depression angles in this section. To be specific, SAR images acquired at different depression angles are exploited to generate a support set and query set for few-shot SAR ATR tasks. The detailed experimental settings of MSTAR data are presented in Table IV.

Meta tasks for model training and evaluation are generated according to those implementation steps provided in Fig. 5. Meta-task settings for the meta-training stage and meta-test stage are the same as those of Section III-C. In this section, each experiment is conducted independently for 1000 runs in 1-shot and 5-shot SAR target recognition tasks. Table V lists the minimum recognition accuracy, maximum recognition accuracy, average recognition accuracy, and STD of the proposed method under different experimental settings. The average confusion matrices of recognition results are presented in Table VI.

As can be observed from Table V, regardless of whether the difference in depression angle between the support set

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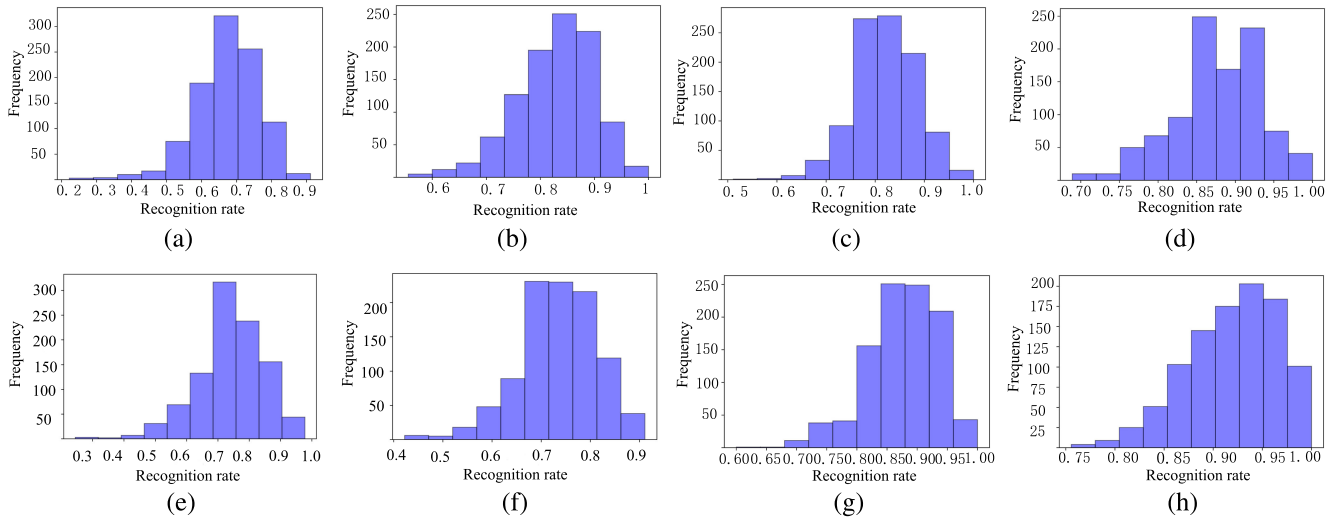


Fig. 7. Statistical histogram of RR for 1000 runs independent experiments. (a) One-shot ProtoNet. (b) Three-shot ProtoNet. (c) Five-shot ProtoNet. (d) Ten-shot ProtoNet. (e) One-shot ACSRNet. (f) Three-shot ACSRNet. (g) Five-shot ACSRNet. (h) Ten-shot ACSRNet.

TABLE IV
EXPERIMENTAL SETTINGS OF MSTAR DATASET

No.	Settings	Depression		
		Training	Test	
			S	Q
1	15s15q	17°	15°	15°
2	17s15q		17°	15°
3	17s17q		17°	17°
4	17s30q		17°	30°
5	30s30q		30°	30°

TABLE V
EXPERIMENTAL RESULTS UNDER FIVE EXPERIMENTAL SETTINGS

No.	Task	Accuracy (%)			STD
		min	max	Avg	
1	1-shot	22.22	97.78	79.55	0.1120
	5-shot	71.11	100	89.77	0.0527
2	1-shot	22.22	95.56	71.46	0.0929
	5-shot	60.00	97.78	81.91	0.0624
3	1-shot	37.78	95.56	75.54	0.0860
	5-shot	66.67	100	87.96	0.0565
4	1-shot	44.44	96.56	70.17	0.0857
	5-shot	60.00	100	82.67	0.0654
5	1-shot	31.11	93.33	70.79	0.0940
	5-shot	57.78	100	81.53	0.0608

and query set is large or small, the proposed method can still achieve an RR of over 70% and 80% for one-shot and five-shot task, respectively, which demonstrates the effectiveness of the proposed method. In addition, it is worth noting that the minimum recognition accuracy can go up to 70% for a five-shot ATR task in 15s15q experimental settings. From the experimental results in Table VI, one can see that the recognition accuracy on each category of the target can be almost equal to the average RR under five experimental settings. These experimental results illustrate that the proposed method has balanced recognition performance in three categories of target recognition tasks.

E. Contrastive Experiments

To verify the superiority of the proposed method, several state-of-the-art few-shot methods, including prototypical network (ProtoNet) [49], relation network (RelationNet) [26], meta-learning SAR (MSAR) [33], BSCapNet [38], and MGA-Net [37], are employed as competitors to conduct multiple contrastive experiments in this section. The experimental settings are consistent with Section III-D. All experiments are conducted independently for 1000 runs, and the average recognition accuracy of each method is listed in Table VII. From the experimental results in Table VII, it can be apparently observed that the proposed method performs better than competitors in different experimental settings for one-shot and five-shot ATR tasks. Therefore, we can state that our proposed method has huge potential in solving few-shot SAR ATR tasks compared with some state-of-the-art.

F. Effectiveness Analysis

In this section, a series of experiments are designed to investigate the effectiveness of the key components of the proposed method and the impact of hyperparameter values on the recognition performance of the model. First, ablation experiments are performed to explore the efficacy of the siamese feature embedding network and few-shot subspace classification module in the proposed method. Then, the enumeration method is exploited to investigate the influence of different values of λ

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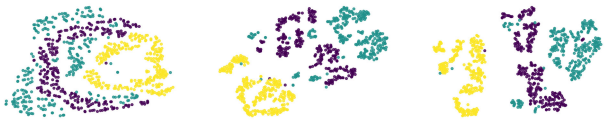
TABLE VI
AVERAGE CONFUSION MATRICES OF RECOGNITION RESULTS UNDER FIVE EXPERIMENTAL SETTINGS

Task		3-way 1-shot				3-way 5-shot			
	Real \ Predicted	2S1	BRDM2	ZSU234	Avg(%)	2S1	BRDM2	ZSU234	Avg(%)
15s15q	2S1	**78.29**	10.51	11.20	79.55	**89.64**	5.24	5.12	89.77
	BRDM2	9.53	**80.90**	9.57		4.81	**89.72**	5.47	
	ZSU234	10.21	10.31	**79.48**		4.74	5.31	**89.95**	
17s15q	2S1	**71.25**	14.40	14.35	71.46	**83.14**	9.03	8.83	83.08
	BRDM2	14.47	**71.63**	14.17		8.59	**83.42**	7.99	
	ZSU234	14.35	14.07	**71.78**		8.10	9.21	**82.69**	
17s17q	2S1	**75.59**	12.51	11.90	75.54	**88.14**	5.48	6.38	87.96
	BRDM2	11.90	**75.45**	12.65		5.57	**88.15**	6.28	
	ZSU234	12.75	11.66	**75.59**		6.33	6.08	**87.59**	
17s30q	2S1	**70.96**	14.54	14.50	70.17	**82.25**	8.61	9.14	82.67
	BRDM2	14.51	**69.99**	15.50		8.10	**82.75**	9.15	
	ZSU234	15.95	14.48	**69.57**		8.87	8.41	**82.72**	
30s30q	2S1	**69.63**	14.78	15.59	70.79	**81.21**	9.56	9.23	81.53
	BRDM2	14.71	**71.26**	14.03		9.36	**81.64**	9.00	
	ZSU234	13.25	15.27	**71.49**		9.06	9.19	**81.75**	

and γ on the recognition performance of the proposed method. Eventually, we design an experiment to investigate the impact of the subspace dimension on recognition performance. In this section, SAR images for three categories of ground military targets, such as 2S1, BRDM2, and ZSU234, are utilized to simulate a three-way five-shot SAR target recognition task, and all experiments are conducted in the 17s30q experiment setting. To get statistically significant results, each experiment is employed independently for 1000 runs.

1) Ablation Experiment: The proposed method consists of a feature embedding network based on the siamese architecture and a few-shot subspace classifier. Considering that the basic architecture of the proposed method is similar to ProtoNet, we can add the feature embedding network and few-shot subspace classification module to ProtoNet, respectively, to investigate the efficacy of the two key components in improving target recognition performance. First, we qualitatively analyze the efficacy of the feature embedding network of the proposed method in improving target recognition performance from the perspective of sample feature distribution. Specifically, this experiment brings to the τ-SNE tool [53] in the Python package to visualize original SAR images, ProtoNet embedding features, and our embedding features, respectively. Fig. 8 displays three types of embedding distributions.

One can see from Fig. 8(a) that the distribution of original images within a class is scattered, and there are serious overlaps between classes. From Fig. 8(b) and (c), it is clear that



(a) (b) (c)

Fig. 8. τ-SNE visualization results. (a) Original SAR images. (b) ProtoNet. (c) Ours ACSRNet.

the experimental results of both ProtoNet and our proposed method show better separability between classes than those shown in Fig. 8(a). Besides, it can be observed that our proposed method can gain a discriminative feature space with intraclass compactness and interclass divergence compared with ProtoNet. It follows that our proposed method can improve the performance of the model by enhancing the discrimination of feature space.

Below, we quantitatively investigate the efficacy of a feature embedding network with dynamic-aware convolution, a few-shot subspace classifier, and dual-loss in improving the recognition performance of the proposed method. For the sake of simplicity, ProtoNet with dynamic-aware convolution is termed variant-1, ProtoNet with few-shot subspace classifier is termed variant-2, and the proposed method without dual-loss is named variant-3. Table VIII lists the RR of each method. From experimental results in Table VIII, one can see that each key component of the proposed method plays a key role in improving few-shot target recognition performance.

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TABLE VII
RECOGNITION PERFORMANCE COMPARISON UNDER FIVE EXPERIMENTAL SETTINGS

Task	Method	Experiment Settings				
		15s15q	17s15q	17s17q	17s30q	30s30q
3-way 1-shot	ProtoNet [49]	71.51	66.81	69.90	64.05	66.76
	RelationNet [26]	74.22	67.50	69.95	64.41	62.78
	MSAR [33]	67.71	66.59	68.43	63.71	64.51
	BSCapNet [38]	72.01	70.84	74.37	73.98	67.22
	MGA-Net [37]	76.49	71.28	73.25	73.16	69.86
	Our ACSRNet	**79.55**	**74.58**	**75.54**	**74.02**	**70.79**
3-way 5-shot	ProtoNet [49]	80.28	81.29	82.22	78.24	78.59
	RelationNet [26]	79.54	80.46	81.21	74.36	74.60
	MSAR [33]	82.62	82.21	83.83	77.89	77.55
	BSCapNet [38]	82.65	78.71	86.60	83.20	80.51
	MGA-Net [37]	87.64	82.21	86.89	82.92	80.63
	Ours ACSRNet	**89.77**	**83.08**	**87.96**	**83.82**	**81.53**

TABLE VIII
ABLATION EXPERIMENTAL RESULTS

Method	ProtoNet	variant-1	variant-2	variant-3	ACSRNet
RR (%)	78.24	80.12	81.29	81.00	83.82

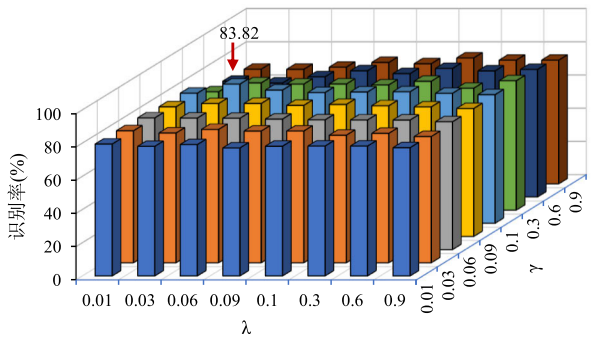


Fig. 9. RR of the proposed method with different values of λ and γ.

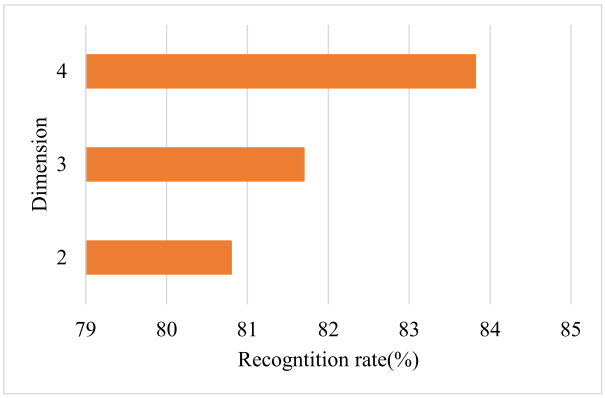


Fig. 10. RR of the proposed method with different subspace dimensions.

2) Sensitivity of Parameter λ and γ : The variables λ and γ are two key parameters that balance the importance of different losses in the dual-loss cost function of the proposed method. We make use of the enumeration method to investigate the influence of different values of the two parameters on the performance of the proposed method. In this experiment, both λ and γ are in the range {0.01, 0.03, 0.06, 0.09, 0.1, 0.3, 0.6, 0.9}. Fig. 9 shows the RR of the proposed method with different values of λ and γ. From Fig. 9, one can see that the RR of the proposed method can slightly fluctuate with the changes in the two parameter values. On the whole, the proposed method can obtain the highest RR when $\lambda = 0.03$ and $\gamma = 0.1$.

3) Selection of Subspace Dimension: The subspace dimension is an important parameter for the few-shot subspace classification module. This experiment investigates the influence of different subspace dimensions on the performance of the proposed method. Because the number of samples for each category of the target is set to 5 in this experiment, the

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subspace dimension ranges from 2 to 4 in this experiment. Fig. 10 presents the RR of the proposed method under different subspace dimensions. As can be observed from Fig. 10, the change of subspace dimension has relatively little effect on the recognition performance of the proposed method. But in view of the RR, the proposed method achieves the best recognition performance when the subspace dimension is 4. Thus, we can state that the discriminative ability of subspace may be enhanced with the increase of subspace dimension in the few-shot SAR ATR task.

IV. Conclusion

This article puts forward a novel few-shot method called ACSRNet to realize SAR target recognition with extremely scarce training samples. The contributions of this article are summarized in three aspects. First, this article proposes a siamese feature embedding network with dynamic-aware convolution to extract discriminative features, aiming to facilitate the identity reasoning of the target. Second, we take advantage of the high-order information of samples to develop a few-shot subspace classification module, which can reason the identity of the unknown target more accurately compared with conventional classification criteria in few-shot SAR target recognition tasks. Third, a dual-loss is proposed to learn a discriminative feature embedding space with intraclass compactness and interclass divergence. Extensive experiments on the MSTAR dataset show that the proposed method for few-shot SAR target recognition is superior to some state-of-the-art.

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Haohao Ren (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2021.

From 2020 to 2021, he was a visiting Ph.D. student with the Remote Sensing Laboratory, Department of Information Engineering and Computer Science, University of Trento, Trento, Italy. He is currently an Assistant Fellow with the School of Information and Communication Engineering, UESTC. His research interests include radar signal processing, synthetic aperture radar (SAR), target recognition, pattern recognition, deep learning, and Bayesian learning.



Xuelian Yu (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2008.

From 2008 to 2010, she held a post-doctoral position at the Department of Electronic Engineering, UESTC. From 2014 to 2015, she was a Visiting Scholar with the Department of Electronic Engineering, The Ohio State University, Columbus, OH, USA. Since 2010, she has been an Associate Professor with the School of Information and Communication Engineering, UESTC. Her research interests include radar signal processing, synthetic aperture radar, target recognition, target tracking, and manifold learning.



Sen Liu (Student Member, IEEE) received the B.S. degree from the School of Electronics and Information Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2020, where he is currently pursuing the master's degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar, automatic target recognition, and deep learning.



Lin Zou (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in signal and information processing from the School of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1999, 2002, and 2013, respectively.

From 2014 to 2015, he was a Visiting Scholar with the Department of Electronic Engineering, Linköping University, Linköping, Sweden. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar signal processing, radar systems, and radar echo simulator.



Xuegang Wang received the B.S., M.S., and Ph.D. degrees in signal and information processing from the School of Electronic Engineering, Xidian University, Xi'an, China, in 1984, 1987, and 1992, respectively.

From 1992 to 1994, he held a post-doctoral position at the Department of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), where he has been a Professor since 2000, Chengdu, China. He is currently the Director of the Laboratory of Radar System and Digital Technology. His research interests include radar signal processing, target recognition and detection, radar echo simulator, and statistical signal processing.



Hao Tang received the master's degree from the School of Electronics and Computer Engineering, Peking University, Beijing, China, in 2016, and the Ph.D. degree from the Multimedia and Human Understanding Group, University of Trento, Trento, Italy, in 2021.

He was a Visiting Scholar with the Department of Engineering Science, University of Oxford, Oxford, U.K. He is currently a Post-Doctoral Research Fellow with the Computer Vision Laboratory, ETH Zurich, Zurich, Switzerland. His research interests are deep learning, machine learning, and their applications to computer vision.

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